Rotaract Club of Xiamen University Malaysia

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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Chili Pan MEE Penang
Selected Domain	Domain 2 - Alpha Red
Selected SDG Track (1–17)	SDG 8 - Decent Work And Economic Growth
Project / Solution Name	Liggy
Name (Leader)	Tai Kai Xuan
Email & Phone (Leader)	elisebelletai1228@gmail.com, 012-297 3380
University Affiliation(s)	Taylor's University, University Teknologi Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

In Malaysia, micro and small businesses (SMEs) face operational issues because they rely heavily on manual processes. Instead of focusing on growing their business, owners must handle extra tasks, including tracking down missed or incorrect invoice details and tracing payments across multiple platforms. This significantly delays production and productivity.

Recently, Malaysia announced plans to increase the minimum wage, jumping from RM1,700 to RM2,500 by 2027. Due to a tighter budget, this makes smaller enterprises hesitate even more about whether to hire dedicated admins just to handle repetitive data entry. Even worse, the majority of SMEs don't have sufficient digital literacy to adopt complex software that requires costly training sessions.

To survive in the market, micro-enterprises are forced to sacrifice their limited free time and capital, along with carrying excessive loans from various third parties. If this doesn't get resolved, it could lead to rising national bankruptcy rates. Besides the social impact, too much reliance on paper-based record-keeping indicates that more hard-copy invoices will be printed out. Consequently, lots of natural resources will be wasted when the workflow should have been fully digitised.

By providing a dedicated innovation solution, we not only help SMEs ensure a stable and positive cash flow, but also contribute to a greener environment. This would align with SDG 8 (Decent Work and Economic Growth) and SDG 9 (Industry, Innovation, and Infrastructure).

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Liggy is a unified e-invoicing web application built to minimise the heavy reliance on manual processing that is constantly draining resources. At its core, the solution involves a Python-based multi-agent AI workflow that acts as a smart invoicing assistant. With Liggy, we designed the flexibility to handle invoices: users can manually type in invoice details or upload unstructured documents for automated analysis. Liggy supports various file formats, including PDF, JPEG, JSON, TXT, and CSV files.

Once the document is uploaded, Liggy extracts the relevant billing details, cutting down the time used for data entry. It doesn't just detect the presence of data; it also runs localized validations to ensure data accuracy, such as verifying that a buyer's Tax Identification Number (TIN) matches the expected Malaysian format. Additionally, not every execution is automated. It puts the extracted data next to the live PDF review. SMEs can review and edit the draft before approving and saving the invoice, or rejecting it.

Once approved, the user can browse the previously generated e-invoices, check whether the invoices are typed or AI-extracted, and send the invoices to customers via email. Customer profiles are automatically updated for calculating cumulative spend and past transactions. We also prepared the features of "Daily Sales Report" and "Monthly Insights" to analyse past performance. Liggy also shows users several key insights, such as identifying their highest-value transactions. To help users navigate Liggy, we also implemented "Liggy AI" for them to ask queries regarding application guidance.

We apply the knowledge from Phase 1 workshops to our solutions:

RunPod & GenAI APIs: Host the model inference endpoint powering the Python agent's document analysis.

Replit: Facilitates rapid UI iteration for our React/Vite frontend.

Database Architecture: Connects our Node.js/Express backend to Firebase Cloud Firestore for secure, scalable storage.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Technology Stack & Architecture

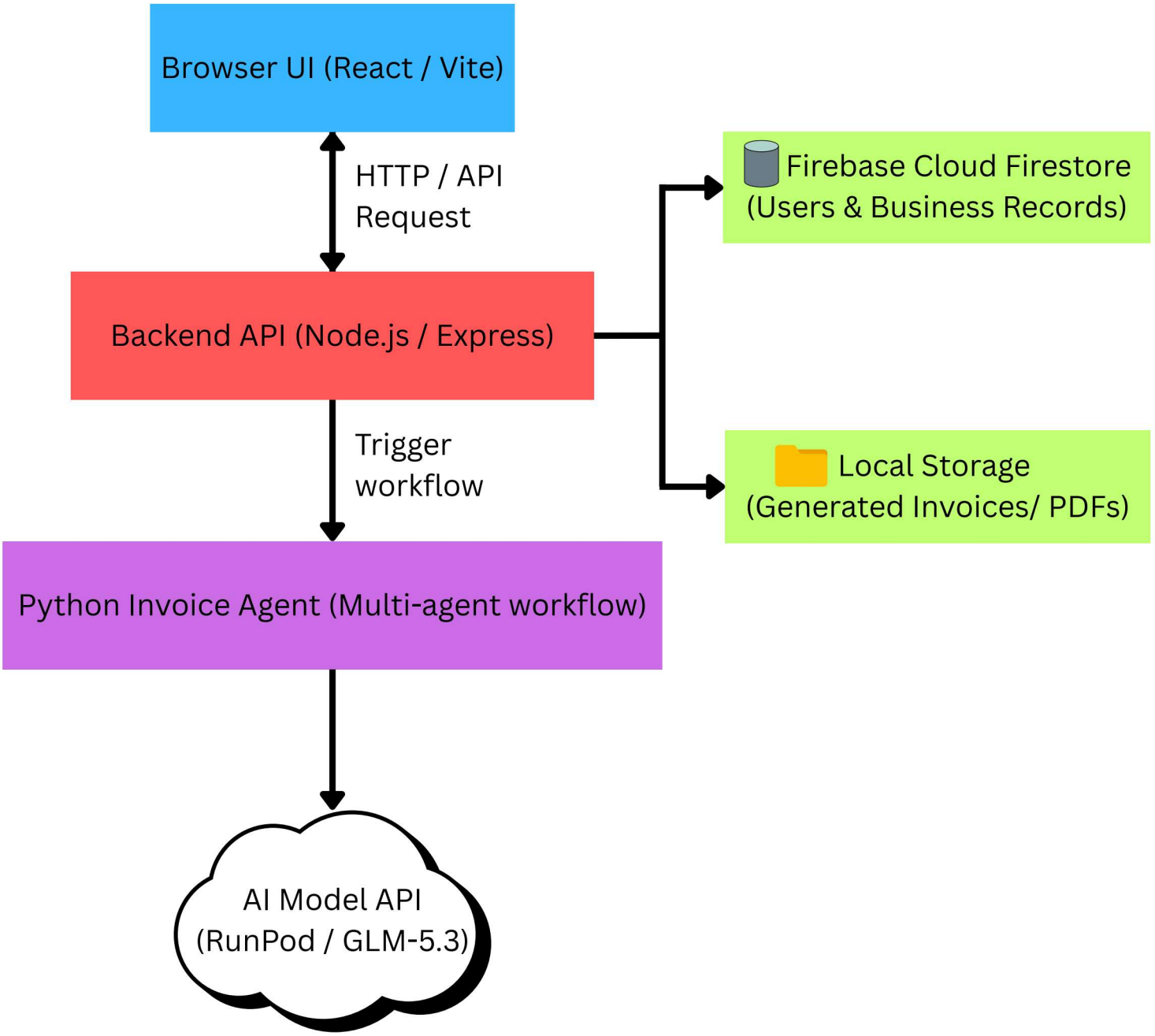
Liggy follows a simple, decoupled flow (refer to the following page for the attached System Block Diagram appendix). Users interact with the frontend React/Vite, which sends HTTP requests to a Node.js/Express backend. Firebase Cloud Firestore is used to manage user and business details, while local storage helps in managing generated e-invoices in PDF format. For intelligent processing, the backend triggers a Python-based LangGraph multi-agent workflow.

SDC Sponsor Integration

Yes, we use the RunPod Cloud Platform and API Service to host and access the GLM-5.3 model. By integrating RunPod's endpoints through the standard OpenAI library, Liggy has the architecture to run both the interactive chatbot and automated document processing.

Conversational AI: Processes user queries via API requests to the GLM-5.3 model on RunPod, where the system receives the JSON payload and converts it to natural, human-readable language.

Multi-Agent Invoicing: RunPod endpoints drive the agent pipeline to analyse uploaded unstructured documents in JPG or PDF format, extract invoice details into a temporary data structure, and validate the details prior to final PDF generation.


Browser UI (React / Vite)
HTTP / API Request
Backend API (Node.js / Express)
Firebase Cloud Firestore (Users & Business Records)
Local Storage (Generated Invoices/ PDFs)
Trigger workflow
Python Invoice Agent (Multi-agent workflow)
AI Model API (RunPod / GLM-5.3)

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

When we were developing Liggy, we realised that most of the SMEs come from various technical backgrounds, and not everyone has high digital literacy. To ensure universal access to Liggy, we built "Liggy AI". Unlike a pre-written manual, Liggy AI acts as an interactive guide. Users can use human language for queries, and they will be provided with specific step-by-step instructions.

Before publishing Liggy to the public, we make sure that Liggy actually works with real-world examples. For instance, we used our own email address to verify the function of delivering the generated e-invoice via SMTP. Additionally, we run on both manual typing inputs and AI extraction tools. This ensures that every customer record and generated PDF are secure.

To prepare Liggy for Phase 2, we attempted to enhance its data security by inserting all database URIs and external API keys through production Environment Variables instead of GitHub. Beyond security, we will utilise the Phase 2 resources and credit to upgrade "Liggy AI". Currently, it only answers basic and predictable questions that are related to the web application features. In the future, we aim to train the model to handle more complex and flexible business queries.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 8.2: "Achieve higher levels of economic productivity through diversification, technological upgrading and innovation, including through a focus on high-value added and labour-intensive sectors."
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	The traditional method of processing invoices in SMEs generates approximately 50g of CO2e per physical paper invoice and requires an average of 15 minutes per document for data entry and processing. As a result, this increases resource waste, prolongs payment cycles, and harms the environment.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	1. 80% reduction in average time taken in time taken to process an invoice. 2. 40% reduction in overall operational costs, tracking savings from reduced administrative overhead, lowered recruitment needs, and mitigated human errors. 3. 80% reduction in volume of physical paper waste.

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	By Phase 2, our system aims to achieve a 100% elimination of physical paper for all invoices processed through Liggy and reduce the average time taken to process an invoice from 15 minutes to 3 minutes only. The prototype will automatically track and highlight business patterns. This includes identifying which clients have the highest transaction frequency, which generate the highest pay per transaction, and which businesses retain the most customers.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
OpenAI Python Library	Open-Source Library	Apache License 2.0
LangGraph Framework	Open-Source Framework	MIT License
RunPod (GLM-5.3 Model)	Pre-Trained AI Model	GLM Model License
Google Gemini	Pre-Trained AI Model	Google API Terms of Service

6.3 Employment-Related IP Conflicts

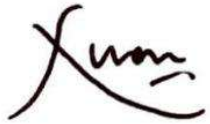
Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



24 October 2026